

Requirement Analysis Phase

Data Flow Diagram & User Stories

Date	October 3, 2026
Team ID	SWTID-2026-7210
Project Name	Script Controlled ACL: Restrict record access based on field value
Maximum Marks	4 Marks

1. Data Flow Diagrams (DFD)

A Data Flow Diagram (DFD) maps out the flow of information for any process or system. It uses defined symbols like rectangles, circles, and arrows, plus short text labels, to show data inputs, outputs, storage points, and the routes between each destination. For the Script-Controlled Access Control List (ACL) system, the data flow depicts how an end-user requests a database record, how the system extracts the contextual field attributes, evaluates constraints via the scripted engine, and securely logs the transaction.

DFD Component Breakdown & Architecture Elements

DFD Element Category	Component Name	Data Flow Description / Role in System
External Entity	End-User / Client Application	Initiates CRUD requests for specific database records containing sensitive attributes.
External Entity	Security Administrator	Defines, modifies, and deploys the condition criteria scripts into the central ACL rule engine.
Process Core	1.0 Context Fetch Engine	Intersects data query requests and inspects target record row field values (e.g., status, department, sensitivity level).
Process Core	2.0 Script Evaluation Core	Compiles rules dynamically and maps structural field strings against user session matrices to decide access grants.
Process Core	3.0 Cryptographic Audit Logger	Writes structural verification footprints containing timestamps and runtime logic results to historical stores.
Data Store	D1 - Core Database Records	Holds primary application tables, entity datasets, and

		transactional tables subject to strict access controls.
Data Store	D2 - ACL Script Configurations	Stores authorized operational rules and code blocks written by administrative staff.
Data Store	D3 - Compliance Audit Logs	Maintains tamper-evident sequential trace logs proving regulatory access restrictions.

2. User Stories

User stories describe features from the perspective of the end user or system stakeholder. They articulate what the user wants to achieve and the precise acceptance parameters required to close out development tasks.

User Type	Epic / Requirement	Story ID	User Story / Task Description	Acceptance Criteria	Priority	Release
Security Admin	Rule Management	USN-01	As an Admin, I can write and upload custom access control scripts so that security settings match internal compliance guidelines.	Script editor verifies syntax rules cleanly before deployment.	High	Sprint-1
Security Admin	Dynamic Overrides	USN-02	As an Admin, I can alter conditions based on record fields without applying app downtime hooks.	Modifications reflect across record queries instantly.	High	Sprint-1
End User	Data Queries	USN-03	As an authorized employee, I can view public information columns	Standard query executions run within minimal millisecond	High	Sprint-1

			effortlessly without security obstruction flags.	latency barriers.		
End User	Data Isolation	USN-04	As a restricted team member, I must be blocked from reading records marked 'Confidential' via specific field states.	System throws strict 403 access denial exceptions.	High	Sprint-1
Compliance Auditor	Access Tracking	USN-05	As an Auditor, I want a complete log of script exceptions so I can generate verification evidence for external regulators.	Audit trails capture evaluated field criteria and user attributes securely.	Medium	Sprint-2